

3(a)	<i>any two from:</i> 1 higher / better, resolution A higher resolving power I clearer resolution or greater ability to distinguish between two points / AW ; <i>ignore wavelength values if stated as wavelength but R if stated as resolution values</i> 2 <i>ref. to</i> resolution values ; e.g. able to see points closer together than 200 nm (range 100–300 nm) can see, points up to 0.5 nm (0.0005 µm) apart (range 0.2–1.0 nm) can see structures larger than 0.5 nm 3 thinner sections can be obtained ; A idea that complete image will be in better focus 4 able to see, ribosomes / membranes / detail <u>within</u> organelles ;	2
3(b)	<i>any three from:</i> 1 (shape of) red blood cell is biconcave <i>accept biconcave from a diagram</i> or (shape of) erythroblast is, cuboidal / spherical / not biconcave / irregular / AW ; 2 red blood cell has no nucleus / erythroblast has nucleus ; 3 erythroblast has, mitochondria / organelles / named organelles or red blood cell has no, mitochondria / organelles / named organelles ; 4 red blood cell has more haemoglobin / erythroblast has less haemoglobin ; I <i>ref. to</i> haem <i>statement only about erythroblast:</i> A erythroblast has no haemoglobin <i>statement only about rbc: must say <u>more</u> haemoglobin</i> <i>statement with both:</i> A red blood cell has haemoglobin, erythroblast has no haemoglobin 5 AVP ; e.g. erythroblast, is larger / basophilic red blood cell is more flexible	3

Question	Answer	Marks														
3(c)(i)	3 structures correct and involvement, incorrect / not stated, allow 1 mark three correct rows I extra rows added with additional structures <table><tr><td>nucleus / chromosome</td><td>(has) gene / DNA, coding for protein or (for) transcription / mRNA synthesis ; A produces ribosomes if stated as made in nucleolus</td></tr><tr><td>nucleolus</td><td>produces, rRNA / ribosomes / ribosomal subunits ;</td></tr><tr><td>ribosome A ribosomal subunit</td><td>(site of) polypeptide / protein, synthesis A described I makes amino acids A to synthesise enzymes or (for) translation or binding of, mRNA / tRNA ;</td></tr><tr><td>rough endoplasmic reticulum A rough ER / RER R wrong word for 'rough' if RER also stated</td><td>site of, polypeptide / protein, synthesis A described A to synthesise enzymes or (for) translation or (for) attachment of ribosomes or protein / post-translational, modification A examples or protein transport ; I packaging proteins</td></tr><tr><td>transport vesicle</td><td>to move protein from RER to Golgi (body / apparatus / complex) ;</td></tr><tr><td>Golgi (body / apparatus / complex)</td><td>for, protein / post-translational, modification / AW ; A examples I packaging proteins</td></tr><tr><td>mitochondrion</td><td>provides / produces, ATP for, tRNA aminoacylation / charging amino acids before attachment to tRNA ;</td></tr></table>	nucleus / chromosome	(has) gene / DNA, coding for protein or (for) transcription / mRNA synthesis ; A produces ribosomes if stated as made in nucleolus	nucleolus	produces, rRNA / ribosomes / ribosomal subunits ;	ribosome A ribosomal subunit	(site of) polypeptide / protein, synthesis A described I makes amino acids A to synthesise enzymes or (for) translation or binding of, mRNA / tRNA ;	rough endoplasmic reticulum A rough ER / RER R wrong word for 'rough' if RER also stated	site of, polypeptide / protein, synthesis A described A to synthesise enzymes or (for) translation or (for) attachment of ribosomes or protein / post-translational, modification A examples or protein transport ; I packaging proteins	transport vesicle	to move protein from RER to Golgi (body / apparatus / complex) ;	Golgi (body / apparatus / complex)	for, protein / post-translational, modification / AW ; A examples I packaging proteins	mitochondrion	provides / produces, ATP for, tRNA aminoacylation / charging amino acids before attachment to tRNA ;	3
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3(c)(ii)	carbonic anhydrase ; R carbon anhydrase / anhydrase	1														

Question	Answer	Marks
3(d)	<i>assume spherocytosis type 2 unless stated otherwise</i> <i>any two from:</i> spherical means reduced surface area (to volume ratio) (so less oxygen diffuses in) ; A SA for surface area <i>idea of</i> further distance for oxygen to reach, (some) haemoglobin molecules / centre of red blood cell ; not enough time to reach, same level of / 98%, saturation of haemoglobin ; <i>idea of blood flowing through</i> <i>ref. to</i> spherical shape / larger / less flexible, so less able to pass through, (pulmonary capillary) network / AW ;	2
3(e)	<i>any three from:</i> <i>(water potential gradient created / loss of equal water potentials, so)</i> <i>similarities</i> 1 water enters cells, by osmosis / down the water potential gradient / from high to low water potential ; A Ψ for water potential 2 both type of cell, swell / increase in size ; <i>differences</i> 3 spherocytosis cells, burst / lyse, more easily / before red blood cells ; A spherocytosis cells burst more quickly A spherocytosis cells burst and red blood cells do not burst 4 AVP ; e.g. spherocytosis cells already more swollen spherocytosis cells have more pressure exerted (on their cell surface membrane) than normal red blood cells AW <i>idea of</i> spherocytosis cells unable to take in as much water and stay intact <i>ref. to</i> less stable / less support for membrane, so weaker membrane (for spherocytosis cells) I less flexible	3